Activity: probabilities

Group members:

Putting data

Let's revisit the putting data. Our regression model is

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{ length}, \qquad \text{or equivalently} \qquad \pi = \frac{\exp\{\beta_0 + \beta_1 \text{ length}\}}{1 + \exp\{\beta_0 + \beta_1 \text{ length}\}},$$

where $\pi = P(\text{make the putt})$.

Suppose you attempt 15 putts, and observe the following outcomes:

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

1. What is the probability of the observed data if $\beta_0 = 3$ and $\beta_1 = -0.5$?

2. What is the probability of the observed data if $\beta_0 = 4$ and $\beta_1 = -0.75$?